

The method of determining the standard matrices of the Mueller calculus and the manner of using the matrices have been discussed in Chapter 8. The same applies to the standard matrices of the Jones calculus.

The commonly used matrices are listed in the present appendix. Obviously, the exact forms of the matrices (for example, the signs of terms involving the sine of an angle) depend on the conventions used in describing the light, the polarizers, and the retarders. Accordingly, the forms listed below are valid only with respect to the conventions used in this book. See Chapters 1, 2, 3, and 7.

Abbreviations. The symbols C_1 and S_1 stand for $\cos \theta$ and $\sin \theta$ for polarizers, and for $\cos \rho$ and $\sin \rho$ for retarders. The symbols C_2 and S_2 stand for $\cos (2\theta)$ and $\sin (2\theta)$ for polarizers and for $\cos (2\rho)$ and $\sin (2\rho)$ for retarders. Also,

$$\begin{aligned} D &= M \sin \tfrac{1}{2}\delta, \\ E &= C \sin \tfrac{1}{2}\delta, \\ F &= S \sin \tfrac{1}{2}\delta, \\ G &= \cos \tfrac{1}{2}\delta, \end{aligned}$$

where M , C , and S are the second, third, and fourth Stokes parameters of the normalized fast eigenvector of a given retarder, and δ (radians or degrees) is the retardance. Finally,

$$\begin{aligned} Y &= \cos (2 \arctan b/a), & P &= e^{i\pi/4}, \\ Z &= \pm \sin (2 \arctan b/a), & Q &= e^{-i\pi/4}. \end{aligned}$$

In the expression for Z the positive or negative sign applies according as the sense of the sectional pattern of the major eigenvector is clockwise or counterclockwise.

Standard Matrices. The standard matrices of miscellaneous devices, polarizers, and retarders are listed below.

Optical device	Mueller matrix	Jones matrix
<i>Miscellaneous devices</i>		
Region devoid of matter; or ideal plate of isotropic, nonabsorbing glass	$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$	$\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$
Ideal plate of isotropic, absorbing glass whose transmittance is k or p^2	$\begin{bmatrix} k & 0 & 0 & 0 \\ 0 & k & 0 & 0 \\ 0 & 0 & k & 0 \\ 0 & 0 & 0 & k \end{bmatrix}$	$\begin{bmatrix} p & 0 \\ 0 & p \end{bmatrix}$
Totally absorbing plate	$\begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$	$\begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$
Ideal depolarizer (nothing closely resembling such a device exists)	$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$	None

Ideal homogeneous linear polarizer

Azimuth θ of transmission axis

0° —	$\frac{1}{2} \begin{bmatrix} 1 & 1 & 0 & 0 \\ 1 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$	$\begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$
90° 	$\frac{1}{2} \begin{bmatrix} 1 & -1 & 0 & 0 \\ -1 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$	$\begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}$
45° /	$\frac{1}{2} \begin{bmatrix} 1 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \\ 1 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$	$\frac{1}{2} \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$

$$\begin{array}{c} -45^\circ \\ \diagdown \end{array} \quad \frac{1}{2} \begin{bmatrix} 1 & 0 & -1 & 0 \\ 0 & 0 & 0 & 0 \\ -1 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \quad \frac{1}{2} \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix}$$

$$\text{General} \quad \frac{1}{2} \begin{bmatrix} 1 & C_2 & S_2 & 0 \\ C_2 & C_2^2 & C_2 S_2 & 0 \\ S_2 & C_2 S_2 & S_2^2 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \quad \begin{bmatrix} C_1^2 & C_1 S_1 \\ C_1 S_1 & S_1^2 \end{bmatrix}$$

Ideal homogeneous nonlinear polarizer

$$\begin{array}{c} \text{Right circular} \\ \bigcirc \end{array} \quad \frac{1}{2} \begin{bmatrix} 1 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 1 \end{bmatrix} \quad \frac{1}{2} \begin{bmatrix} 1 & -i \\ i & 1 \end{bmatrix}$$

$$\begin{array}{c} \text{Left circular} \\ \bigcirc \end{array} \quad \frac{1}{2} \begin{bmatrix} 1 & 0 & 0 & -1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ -1 & 0 & 0 & 1 \end{bmatrix} \quad \frac{1}{2} \begin{bmatrix} 1 & i \\ -i & 1 \end{bmatrix}$$

$$\begin{array}{c} \text{Right elliptical} \\ \text{---} \circ \text{---} \\ \theta = 0^\circ \\ b/a = 0.5 \end{array} \quad \frac{1}{2} \begin{bmatrix} 1 & 0.6 & 0 & 0.8 \\ 0.6 & 0.36 & 0 & 0.48 \\ 0 & 0 & 0 & 0 \\ 0.8 & 0.48 & 0 & 0.64 \end{bmatrix} \quad \frac{1}{2} \begin{bmatrix} 2 & -i \\ i & 0.5 \end{bmatrix}$$

$$\begin{array}{c} \text{Right elliptical} \\ \text{---} \circ \text{---} \\ \theta = 22.5^\circ \\ b/a = 0.318 \end{array} \quad \frac{1}{2} \begin{bmatrix} 1 & \sqrt{\frac{1}{3}} & \sqrt{\frac{1}{3}} & \sqrt{\frac{1}{3}} \\ \sqrt{\frac{1}{3}} & \frac{1}{3} & \frac{1}{3} & \frac{1}{3} \\ \sqrt{\frac{1}{3}} & \frac{1}{3} & \frac{1}{3} & \frac{1}{3} \\ \sqrt{\frac{1}{3}} & \frac{1}{3} & \frac{1}{3} & \frac{1}{3} \end{bmatrix} \quad 0.288 \begin{bmatrix} 2.73 & 1-i \\ 1+i & 0.733 \end{bmatrix}$$

$$\begin{array}{c} \text{Elliptical,} \\ \text{general case} \end{array} \quad \frac{1}{2} \begin{bmatrix} 1 & C_2 Y & S_2 Y & Z \\ C_2 Y & C_2^2 Y^2 & C_2 S_2 Y^2 & C_2 Y Z \\ S_2 Y & C_2 S_2 Y^2 & S_2^2 Y^2 & S_2 Y Z \\ Z & C_2 Y Z & S_2 Y Z & Z^2 \end{bmatrix} \quad \begin{bmatrix} |m|^2 & mn^* \\ m^* n & |n|^2 \end{bmatrix}$$

where the
Jones major
and minor
eigenvectors
are

$$\begin{bmatrix} m \\ n \end{bmatrix} \quad \text{and} \quad \begin{bmatrix} -n^* \\ m^* \end{bmatrix}$$

	<i>Other polarizers</i>			
Homogeneous, non-scattering, nondepolarizing, nonbirefringent polarizer with principal transmittances $k_1 = p_1^2$ and $k_2 = p_2^2$ and with the transmission axis horizontal ($\theta = 0^\circ$)	$\frac{1}{2}$	$\begin{bmatrix} k_1 + k_2 & k_1 - k_2 & 0 & 0 \\ k_1 - k_2 & k_1 + k_2 & 0 & 0 \\ 0 & 0 & 2\sqrt{k_1 k_2} & 0 \\ 0 & 0 & 0 & 2\sqrt{k_1 k_2} \end{bmatrix}$	$\begin{bmatrix} p_1 & 0 \\ 0 & p_2 \end{bmatrix}$	
Same but with $\theta = 90^\circ$	$\frac{1}{2}$	$\begin{bmatrix} k_1 + k_2 & -k_1 + k_2 & 0 & 0 \\ -k_1 + k_2 & k_1 + k_2 & 0 & 0 \\ 0 & 0 & 2\sqrt{k_1 k_2} & 0 \\ 0 & 0 & 0 & 2\sqrt{k_1 k_2} \end{bmatrix}$	$\begin{bmatrix} p_2 & 0 \\ 0 & p_1 \end{bmatrix}$	
Inhomogeneous right-circular polarizer consisting of two ideal homogeneous layers: linear polarizer with horizontal axis and linear retarder with retardance of 90° and fast axis at 45° ; light assumed to be incident on linearly polarizing layer	$\frac{1}{2}$	$\begin{bmatrix} 1 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 \end{bmatrix}$	$\frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 0 \\ i & 0 \end{bmatrix}$	

Ideal homogeneous linear retarder with retardance $\delta = 90^\circ$

Azimuth ρ of fast axis

0° —	$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & -1 & 0 \end{bmatrix}$	$\begin{bmatrix} e^{i\pi/4} & 0 \\ 0 & e^{-i\pi/4} \end{bmatrix}$
90° 	$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & -1 \\ 0 & 0 & 1 & 0 \end{bmatrix}$	$\begin{bmatrix} e^{-i\pi/4} & 0 \\ 0 & e^{i\pi/4} \end{bmatrix}$
45° /	$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & -1 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \end{bmatrix}$	$\frac{1}{\sqrt{2}} \begin{bmatrix} 1 & i \\ i & 1 \end{bmatrix}$

$$\begin{array}{c}
 -45^\circ \\
 \diagdown
 \end{array}
 \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \\ 0 & -1 & 0 & 0 \end{bmatrix}
 \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & -i \\ -i & 1 \end{bmatrix}$$

$$\text{General} \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & C_2^2 & C_2 S_2 & -S_2 \\ 0 & C_2 S_2 & S_2^2 & C_2 \\ 0 & S_2 & -C_2 & 0 \end{bmatrix}
 \begin{bmatrix} C_1^2 P + S_1^2 Q & \sqrt{2} i C_1 S_1 \\ \sqrt{2} i C_1 S_1 & C_1^2 Q + S_1^2 P \end{bmatrix}$$

Ideal homogeneous linear retarder with retardance $\delta = 180^\circ$

Azimuth ρ of fast axis

$$\begin{array}{c}
 0^\circ \text{ or } 90^\circ \\
 \\
 \pm 45^\circ
 \end{array}
 \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & -1 \end{bmatrix}
 \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & -1 \end{bmatrix}
 \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$$

$$\text{General} \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & C_2^2 - S_2^2 & 2C_2 S_2 & 0 \\ 0 & 2C_2 S_2 & S_2^2 - C_2^2 & 0 \\ 0 & 0 & 0 & -1 \end{bmatrix}
 \begin{bmatrix} C_2 & S_2 \\ S_2 & -C_2 \end{bmatrix}$$

Other ideal homogeneous linear retarders

$$\delta = 360^\circ, \text{ any } \rho
 \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}
 \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$\text{Any } \delta, \rho = 0^\circ
 \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & D^2 + G^2 & 0 & 0 \\ 0 & 0 & -D^2 + G^2 & 2DG \\ 0 & 0 & -2DG & 2G^2 - 1 \end{bmatrix}
 \begin{bmatrix} e^{i\delta/2} & 0 \\ 0 & e^{-i\delta/2} \end{bmatrix}$$

$$\text{Any } \delta, \text{ any } \rho$$

Mueller matrix

$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & D^2 - E^2 + G^2 & 2DE & -2EG \\ 0 & 2DE & -D^2 + E^2 + G^2 & 2DG \\ 0 & 2EG & -2DG & 2G^2 - 1 \end{bmatrix}$$

$$\begin{array}{c} \text{Jones matrix} \\ \begin{bmatrix} C_1^2 e^{i\delta/2} + S_1^2 e^{-i\delta/2} & C_1 S_1 2i \sin \frac{1}{2}\delta \\ C_1 S_1 2i \sin \frac{1}{2}\delta & C_1^2 e^{-i\delta/2} + S_1^2 e^{i\delta/2} \end{bmatrix} \end{array}$$

or

$$\begin{bmatrix} m_{11} & m_{12} \\ m_{21} & m_{22} \end{bmatrix}$$

where

$$m_{11} = \cos^2 \rho + e^{-i\delta} \sin^2 \rho,$$

$$m_{22} = \sin^2 \rho + e^{-i\delta} \cos^2 \rho,$$

$$m_{12} = m_{21} = (1 - e^{-i\delta}) \cos \rho \sin \rho$$

Ideal homogeneous nonlinear retarders

$$\begin{array}{l} \text{Right circular,} \\ \delta = 90^\circ \\ (\text{hence } \zeta = 45^\circ) \end{array} \quad \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \quad \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ -1 & 1 \end{bmatrix}$$

$$\begin{array}{l} \text{Left circular,} \\ \delta = 90^\circ \\ (\text{hence } \zeta = -45^\circ) \end{array} \quad \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \quad \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & -1 \\ 1 & 1 \end{bmatrix}$$

$$\begin{array}{l} \text{Right or left} \\ \text{circular,} \\ \delta = 180^\circ \\ (\text{hence } \zeta = \pm 90^\circ) \end{array} \quad \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \quad \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}$$

$$\begin{array}{l} \text{Right circular,} \\ \text{any } \delta \end{array} \quad \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & \cos \delta & \sin \delta & 0 \\ 0 & -\sin \delta & \cos \delta & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \quad \begin{bmatrix} \cos \frac{1}{2}\delta & \sin \frac{1}{2}\delta \\ -\sin \frac{1}{2}\delta & \cos \frac{1}{2}\delta \end{bmatrix}$$

$$\begin{array}{l} \text{Left circular,} \\ \text{any } \delta \end{array} \quad \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & \cos \delta & -\sin \delta & 0 \\ 0 & \sin \delta & \cos \delta & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \quad \begin{bmatrix} \cos \frac{1}{2}\delta & -\sin \frac{1}{2}\delta \\ \sin \frac{1}{2}\delta & \cos \frac{1}{2}\delta \end{bmatrix}$$

Elliptical, any δ , any ρ

Mueller matrix

$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & D^2 - E^2 - F^2 + G^2 & 2(DE + FG) & -2(DF + EG) \\ 0 & 2(DE - FG) & -D^2 + E^2 - F^2 + G^2 & 2(DG - EF) \\ 0 & -2(DF - EG) & -2(DG + EF) & -D^2 - E^2 + F^2 + G^2 \end{bmatrix}$$

$$\begin{array}{c} \text{Jones matrix} \\ \left[\begin{array}{cc} C_r^2 e^{i\delta/2} + S_r^2 e^{-i\delta/2} & C_r S_r (2i \sin \frac{1}{2}\delta) e^{-i\gamma} \\ C_r S_r (2i \sin \frac{1}{2}\delta) e^{i\gamma} & C_r^2 e^{-i\delta/2} + S_r^2 e^{i\delta/2} \end{array} \right] \end{array}$$

where $C_r = \cos R$ and $S_r = \sin R$, the quantities R and γ having the significance indicated in Chapters 1 and 2.